INFORMATION SECURITY

PUCIT (FCIT)

NAME: M. AHMAD FARAZ

ROLLNO: BITF24M046

Bad USB / HID Keystroke Injection & Detection

1. Introduction

With the increasing use of USB devices, systems are becoming vulnerable to **Bad USB / HID injection attacks**, where a malicious USB device behaves like a keyboard and executes

automated commands at high speed. Traditional antivirus systems often fail to detect such

attacks because they appear as normal human input.

This project, **HID Biometric Shield**, is designed to detect such threats by analyzing

behavioral biometrics, specifically typing patterns. By monitoring typing speed and rhythm variance, the system distinguishes between human and machine-generated input.

2. Problem Statement

Modern systems cannot easily differentiate between **human typing** and **automated**

keystroke injection from malicious USB devices. This creates a major security loophole

where attackers can execute commands without detection.

The challenge is to develop a system that:

- Detects non-human typing patterns in real time
- Identifies high-speed and highly consistent input behavior
- Alerts users before damage occurs

3. Objectives

The main objectives of this project are:

- To monitor real-time keyboard input behavior
- To calculate typing speed and rhythm variance
- To detect robotic (automated) typing patterns
- To provide visual threat analysis using a GUI dashboard

- To generate alerts and maintain a security audit log
- To simulate attacks for demonstration purposes

4. Methodology / Working

The system works based on **behavioral biometric analysis**:

1. **Data Collection**
   - Captures time intervals between consecutive keystrokes
2. **Feature Extraction**
   - Calculates:
     - Average typing speed
     - Variance in typing rhythm
3. **Threat Analysis**
   - If typing is:
     - Too fast → suspicious
     - Too consistent (low variance) → robotic
   - A threat score is calculated (0–100%)
4. **Decision Making**
   - Low score → Safe
   - Medium score → Suspicious
   - High score → Attack detected
5. **Response**
   - Displays alert message
   - Logs incident in audit file
   - Resets monitoring data

5. Implementation Details

The system is implemented in **Python** using the following technologies:

- **Tkinter** → GUI interface

- **pynput** → Keyboard input monitoring
- **collections.deque** → Efficient storage of keystroke intervals
- **statistics** → Mean and variance calculations
- **threading** → Background input listening

Key Features:

- Real-time biometric dashboard
- Threat probability gauge
- Security audit logging system
- Attack simulation button
- Toggle-based system activation

Core Logic:

- Uses thresholds:
  - Very low variance → robotic input
  - Extremely fast typing → non-human
- Combines both factors to calculate threat percentage

6. Results

The system successfully:

- Differentiates between human and automated typing
- Detects simulated HID injection attacks
- Displays real-time analytics (speed, variance, threat level)
- Generates alerts when threat reaches critical level

During testing:

- Normal human typing → SAFE
- Fast or slightly consistent typing → SUSPICIOUS
- Simulated attack → ATTACK detected

7. Conclusion

The **HID Biometric Shield** provides an effective method for detecting **Bad USB attacks** using behavioral biometrics. Unlike traditional security systems, it focuses on **how input is generated**, rather than what is executed.

This approach enhances system security by:

- Detecting hidden threats
- Providing real-time monitoring
- Preventing unauthorized automated actions

Future improvements may include:

- Machine learning for better accuracy
- Multi-user profiling
- Integration with operating system security